Ysbyty Gwynedd Anaesthetic Department

YG CG 1104: Epidural Analgesia in Adult Acute Pain Management:
Best Practice Guidelines

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To update best practice, and anticoagulants.

Summary

A guideline for epidural analgesia in adults for acute pain management.

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1. Introduction

Epidural analgesia is one of the most effective methods available for the management of acute pain relief after major surgery and has consistently to provide better pain relief than parenteral opioid administration (Schug and Macintyre, 2015).

When local anaesthetics are used with or without opioids, the technique is particularly beneficial for the treatment of pain associated with activities such as coughing or walking, and improving patient outcomes (Manion and Brennan, 2011). Epidural analgesia may lead to reduced post-operative complications (particularly respiratory, cardiac and gastrointestinal) and possibly mortality (Popping et al.,2014).

However, epidural analgesia is also associated with several uncommon but significant complications. Epidural analgesia is initiated and managed by an anaesthetist. To reduce the risk, all medical and nursing staff involved must have a good understanding of this form of pain relief (Macintyre and Schug, 2015).

2. Scope

This guideline is concerned with the management of epidural analgesia in the hospital setting. They are not concerned with the management of epidural analgesia for obstetrics or palliative care within Ysbyty Gwynedd, and intended for all physicians, nurses and pharmacists involved in the management of acute pain.

3. Aims and Objectives

- To manage pain effectively.
- To protect and promote the safety of wellbeing of patients receiving epidural analgesia.
- To provide guidelines for the practitioner undertaking the procedure.
- To promote evidence based and safe practice.
- To share ideas and information with colleagues.

4. Responsibilities

Multidisciplinary Team
• To comply with their own Professional Code of Conduct • To follow local Drug Administration and Intravenous Policies and Guidelines • Ensure professional development regarding epidural analgesia • Be competent in using epidural equipment • Continuously evaluate the effectiveness of the epidural analgesia • Ensure a patients' pain is always managed effectively • Ensure alternative analgesia is prescribed prior to discontinuing epidural analgesia, that the dose is administered, and the effectiveness evaluated, prior to removing the epidural catheter.
Anaesthetist
• Ensure appropriate patient selection and obtain informed consent • Initiate epidural analgesia management and ensure on-going management within a suitable environment • Complete appropriate documentation • Provide clear and explicit prescriptions • Hand over epidural care to appropriately trained medical and nursing staff
Acute Pain Team
• Be available to advise staff regarding epidural analgesia within office hours. Out of hours, contact the on-call anaesthetic team • Provide staff with appropriate education and training • Ensure protocols/guidelines are available for the safe administration of epidural analgesia • Implement preventive maintenance for equipment • Evaluate the management of pain (e.g. via audit) • Develop the epidural service is accordance with evidence-based practice • Implement the agreed configuration of epidural devices
Nursing Ward Staff
• Attend epidural management training • Ensure patient education is provided, reinforced and as far as possible understood • Monitor and evaluate patients' pain in accordance with local protocols • Report clinical problems to the Senior Nurse and the Acute Pain Team and/or anaesthetic staff to take appropriate action • Ensure technical faults with equipment are reported to the Acute Pain Team and EBME and that appropriate action is taken • Change bags, infusion rates and give boluses with the prescription limits and according to the patient's condition (if trained and competent) • Once discontinued, remove the epidural catheter (if trained and competent)
Pharmacists
• Ensure an adequate supply of drugs is available for epidural analgesia • Provide prompt, accurate and up-to-date drug information

• Monitor epidural analgesia prescribing
Bio-medical Engineers (EBME)
• Provide support and advise on all infusion devices in conjunction with the Medical Devices Officer. • With the Acute Pain Team, ensure devices are maintained in accordance with manufacturers recommendations.
Head of Nursing and Health Board Management
• Ensure adequate resources are available to provide an epidural analgesia service (e.g. funding for staff, pumps, sterile products, etc.). • Allow adequate study time for staff to attend epidural analgesia training sessions • Promote awareness of these guidelines.

5. Rationale

There are three main advantages to using epidural analgesia:

1. It has the potential to provide dynamic pain relief for many patients (D'Arcy, 2011).

2. The combination of local anaesthetic agents with opioids has a synergistic action that it allows the concentration of each drug to be reduced. This limits unwanted side-effects for each drug (Ballantyne el al., 2015).

3. There is evidence that the use of epidural analgesia may reduce the stress response to surgery or trauma, thereby reducing morbidity, recovery time and hospital stay (Kehlet,1997).

6. Patient selection and consent

1. Patient selection for epidural analgesia should be based on a careful risk/benefit analysis for each patient. Risk factors include impairment of coagulation (pathological or therapeutic); infection; compromised immunity, duration of epidural catheterisation; cardiovascular stability; and inadequate postoperative monitor capability.

2. Continuous epidural analgesia is a significant procedure with specific and potentially serious complications; therefore, informed patient consent should be obtained. The letter informing of the late complications that may occur of

discharge should be given to the patient (Appendix 1) (Faculty of Pain Medicine of the Royal College of Anaesthetists, 2010),

7. Indications

1. Provision of post-operative analgesia for pain resulting from trauma, for example fractured ribs, which may result in respiratory impairment due to pain on breathing (Chapman and Day, 2001).

2. Provision of analgesia during labour (not covered with this policy).

3. An alternative to general anaesthesia, for example, in severe respiratory disease or for patients with malignant hyperthermia (Macintyre and Schug, 2015).

8. Contraindications

Absolute

- Patients with coagulation defects, which may result in haematoma formation and spinal cord compression, for example anticoagulated patients, haemophiliacs, or thrombocytopenia due to disease or as a result or anticancer treatment (Horlocker and Wedel, 1998).
- Systemic sepsis.
- Local sepsis at the site of the proposed epidural injection; the result might be meningitis or epidural abscess formation.
- Proven allergy to the intended drug.
- Unstable spinal fracture.
- Patient refusal to consent to the procedure.

Relative

- Unstable cardiovascular system.
- Spinal deformity
- Raised intracranial pressure (a risk of herniation if a dural tap occurs
- Certain neurological conditions, for example multiple sclerosis, where the epidural may result in an exacerbation of the disease (Hall, 2000)

9. Location of patient receiving epidural analgesia

- Currently, patients receiving epidural analgesia in Ysbyty Gwynedd, should be supported and assessed on ITU, HDU or recovery department.

10. Safe Management of epidural analgesia

- To reduce the risk of intravenous fluids being administered inadvertently via an epidural catheter, or epidural infusion being given intravenously with catastrophic consequences.

- Only dedicated epidural pumps are used.

- Epidural infusions are stored separately from intravenous fluids in a controlled drugs cupboard and colour coded in yellow (NPSA, 2007, Becker et al, 2012).

- Epidural monitoring and observation charts will maximise effectiveness of the epidural, minimise risks, improve recognition and treatment of adverse effects (Appendix 2 and Appendix 3).



11. Education and Competence

- Staff involved with patients receiving epidural analgesia must undertake specific training and should remain updated with refresher sessions.

- All staff using infusion pumps to deliver epidural analgesia must receive appropriate training and assessed as competent.

- A record of training and education should be retained within the Healthboard.

- Clinical Managers must ensure their staff receive adequate epidural analgesia training and refresher session (Dougherty and Lister, 2015).

Pre-procedural considerations

12. Equipment

1. A Tuohy needle (a bevelled curve-up tip needle to reduce the risk of dural puncture, with 1cm length marking, either 16 G or 18 G.

2. An epidural catheter (a relatively stiff catheter made from a polyamide, length 1000mm with clear blue markings completely embedded in the catheter material).

3. A bacterial filter (filters provide an additional degree of safety and control and prevent bacterial infections. Minimal dead space ensures accurate dosing. A high-pressure resistance of up to 7 bar enhances safety during manual injection.

4. A connector (to ensure safe catheter fixation).

5. Loss of resistance device (to aid clear identification of the epidural space).

6. Within Ysbyty Gwynedd the equipment comes in prepared sterile disposable epidural packs.

13. Dressing

1. A transparent moisture-responsive occlusive dressing (e.g. Tegaderm or OPSITE) to allow easy inspection of the site and allow visibility of the catheter distance markings.

2. The epidural should be secured with hypoallergenic tape to the patients back and over their shoulder, to prevent the filter causing a pressure ulcer.

3. The dressing should only be changed using aseptic technique, only is soiled or becomes non-adherent.

4. The epidural site should be inspected every 12 hours (Appendix 2 and Appendix 3).

5. The anaesthetist should document the level of insertion (e.g. L2-L3) the depth to the epidural space and the length of catheter in the epidural space appropriately (usually the anaesthetic chart) (Lister and Dougherty, 2015).

14. Urinary Catheterisation

1. Urinary catheterisation is recommended for patients receiving epidural analgesia.

2. If epidural analgesia is the only indication for urinary catheterisation, it should be removed within 4 hours of removing the epidural catheter. The patient should subsequently be observed for urinary retention.

3. If the patient fails to pass urine 6 hours after removing the urinary catheter, appropriate action should be taken.

15. Drugs used with epidural analgesia

1. The most commonly used solutions for epidural infusions are combinations of low concentrations of local anaesthetics and opioids (Baker el al, 2004, Wheatley et al, 2001). The effects appear to be synergistic and permit reduction in the reduction of each class of drug, while providing better analgesia and lower risk of adverse effects than could be obtained with either agent alone (Curatolo et al, 1998, Fotiadis et al, 2004).

2. Prescriptions for epidural analgesia should be clearly written stating drug(s), concentration (s), diluent and infusion range in accordance with the hospitals medicine policy.

3. The use of premixed bags is recommended to minimise medication errors involving the epidural route (NPSA, 2007). Within Ysbyty Gwynedd the following are available:

   - *0.1% levobupivacaine and 2 micrograms/ml Fentanyl*
   - *0.1% levobupivacaine and 4 micrograms/ml Fentanyl*

4. Oxygen therapy should be prescribed where appropriate.

5. Additional analgesia should be prescribed as necessary.

6. **NO** other opioids should be prescribed whilst the patient is receiving opioid epidural analgesia, unless under the direction of a Specialist Practitioner.

7. Naloxone and an antiemetic should be prescribed to treat the potential side-effects associated with opioids,

8. The epidurals infusion rate should only be prescribed in mls/hr **NOT** in mgs or micrograms/hr (Macintyre and Schug, 2015).

16. Infusion Pump

1. Yellow colour coded infusion lines and dedicated yellow CADD Solis epidural pumps have electronic safeguards to prevent incorrect programming.

2. Clear labelling of epidural catheters and infusion lines is essential.

3. An antibacterial/viral filter located between the epidural catheter and epidural administration set.

4. The epidural pump and infusion bag are locked in a transparent yellow case to avoid patients interfering with the contents.

5. Handbooks should be available to all users of the CADD Solis Pumps

6. The device should allow:
   - Programming in mls/hr
   - Rate capping
   - Printable extended history
   - Locking with restricted access
   - Clear vision of fluids
   - Adjustable pressure limits
   - Bolus administration by specialist users

7. Non-luer lock epidural ports and syringes are currently under discussion.





17. Specific patient preparation

1. Patients undergoing epidural analgesia should always have venous access before the procedure. This is because, although rare, a reaction to the opioid of

local anaesthetic (e.g. respiratory depression or sympathetic blockade) may require immediate access to the venous system.

18. Information and consent prior to using epidural analgesia

The following issues should be discussed with the patient and their families:

- Drug related side effects
- Motor weakness
- Urinary retention
- Risk of temporary or permanent neurological damage

19. Position of epidural

1. Local anaesthetic drugs block nerve fibres at spinal segments adjacent to their site of administration. To ensure the local anaesthetic agents spreads to the dermatome or nerves supplying the area of pain (Appendix 5).

2. The tip of the spinal catheter should be placed within the mid-dermatomal distribution of the pain site:

Surgical site	Catheter location
Thoracic	T6-T9
Upper abdominal	T7-T10
Lower abdominal	T9-L1
Hip/knee	L1-L4

Post-procedural considerations

20. Volume infusion

- The medication ‘spread’ within the epidural space is determined by the site of the epidural catheter, the patient’s age and the volume of the drug being infused (Rockford and DeRuyter, 2009).

- It is therefore important to maintain the hourly infusion rate at a volume which keeps the appropriate nerves blocked.

21. Effectiveness of blockade

- The spinal nerves supply specific areas of the skin known as dermatomes (Appendix 5.)

- Sensitivity to changes in temperature (e.g. ice cold) along the sensory dermatome can be used to assess the level of the epidural block (Appendix 6).

- This levels should be checked to ensure that the epidural is providing is providing pain relief by covering the area of the site of pain but is also necessary to maintain safety. If the block is too high (above T4) there is an increased risk of respiratory and cardiac symptoms.

22. Drug related side-effects

When caring for a patient receiving epidural analgesia, it is important to monitor the patient for the following:

- Drug related side effects
- Pain intensity
- Signs of complication due to the procedure
- Equipment related problems

Opioids

***Respiratory depression*:** due to the action of the opioids on the respiratory centre. Respiratory depression may occur at two different time intervals

- *Early*: usually within 2 hours of the opioid injection. This may occur if high levels of the opioid follow absorption from the epidural space into the systemic circulation (Macintyre and Schug, 2007, 2015).

- *Late*: this may not be seen for 6-12 hours after the opioid has been given. It results from migration of the drug in the CSF to the brainstem and respiratory centre (Macintyre and Schug, 2007, 2015). This is less likely to occur with lipid-soluble opioids such as fentanyl.

***Sedation*:** although there may be several different causes of sedation, epidural opioids can cause sedation owing to their effect on the central nervous system. Opioid induced sedation is often an early warning sign of respiratory depression.

Nausea and vomiting: caused by the action of the opioids on the vomiting centre in the brain stem and stimulation of the chemoreceptor trigger zones in the fourth ventricle in the brain.

Pruritus: although the exact mechanism is unknown, pruritus is presumed to be centrally medicated and a consequence of the activation of opioid receptors in the spinal cord (Sands et al. 1998).

Urinary retention: this is due to opioid inhibition of the micturition reflex which is evoked by increases in bladder volume.

Local anaesthetic agents

Hypotension*:* this can be caused by two mechanisms:

1. Local anaesthetic agents can spread outside the epidural space, blocking the sympathetic nerves. This results in peripheral vasodilation and hypotension. It is more likely to occur if a bolus dose of anaesthetic agent (e.g. 10mL 0.25% bupivacaine) is given to improve pain control (Macintyre and Schug, 2007).

2. If local anaesthetic agent spreads above the T4 dermatome (nipple line), the cardio-accelerator nerves may be come blocked, leading to bradycardia and hypotension (Macintyre and Schug, 2007).

Motor blockade*:* this will depend on the concentration and total dose of the anaesthetic agent and the position of the epidural (Hall, 2000). Motor blockade occurs when the local anaesthetic agent blocks the larger diameter motor nerves. Leg weakness will occur if the motor nerves supplying the legs are blocked.

Urinary retention*: the epidural opioids blockade the nerves supplying the bladder sphincter causing urinary retention.*

Comparison of the possible side effects of epidural opioids and local anaesthetics

	Opioid	**Local anaesthetic**
Respiratory	Delayed opioid-inducted ventilatory impairment (OIVI) Early OIVI	Usually unimpaired
Cardiovascular	Usually no reduction in blood pressure	Overt or postural hypotension Reduced heart rate with high block
Nausea/vomiting	Yes	Less common
Pruritus	Yes	No
Motor	No effect	Block
Sensation	No effect	Block
Urinary retention	Yes	Yes
Gastrointestinal	Decreased motility	Increased motility

23. Patient Observations

1. The frequency and timing of patient observations will depend on the patient's condition. If stable the following are recommended:
    - Blood pressure
    - Pulse
    - Respiratory rate
    - Pain
    - Sedation
    - Oxygen saturations

2. Every 15 minutes for the first 2 hours after insertion of the epidural catheter or at the end of the operation,1 hourly thereafter for the first 24 hours.

3. Unless there is cause for concern, after the first 24 hours following the patients sleep pattern should be disrupted as little as possible.

4. Routine observation, sedation scores and infusion recordings should continue.

5. The epidural infusion device readings should be recorded hourly including:
    - Rate of infusion
    - Total volume of infusion used

6. Two qualified staff must verify the epidural infusion device program and the drug label against the prescription at:
    - Commencement of the infusion
    - Every shift change
    - After any alteration of the infusion

8. This should be documented accordingly (Appendix 2 and Appendix 3).

9. The height of the block should be recorded at least 8 hourly. Ensuring there is no upper limb block helps ensure the block is not too high.

10. The patient's temperature should be recorded at least 4 hourly whilst the epidural catheter is in place.

11. Assessment of the patient's pain should be made:
    - At rest
    - On movement
    - On deep breathing and the worst score recorded

12. The patency of intravenous access should be ensured whilst the epidural is in use.

13. The epidural insertion site should be checked for inflammation, tenderness, leaking and swelling at least once a shift, and following removal daily for 2 days.

24. Management of inadequate analgesia

- If a patient receiving epidural analgesia complains of inadequate pain relief or pain out of proportion to the procedure, initial assessment must also consider the causes of pain, as it may not be the same pain for with the epidural was first commenced (for example, developing peritonitis or compartment syndrome).

- If pain is severe enough to “break though” the epidural analgesia, or the pain is distant for the site not to be covered by the epidural. For example, a trauma patient with a thoracic epidural catheter placed for fractured ribs but has additional injuries at other peripheral sites, additional analgesia may be required (Appendix 8 and Appendix 9).

- If other opioids are needed to supplement epidural analgesia (such as Patent Controlled Analgesia) and specialist advice should be sought from the Acute Pain Team or Consultant Anaesthetist (Macintyre and Schug, 2015)

25. “Step down” analgesia

- Epidural analgesia should be provided for a limited time only to reduce catheter related infections. However, the average duration is 48-72 hours. Epidurals should only be used for longer than 72 hours after careful consideration of the risks versus the benefits for the patient. This decision should be made by a Consultant Anaesthetist.

- Patients and staff need to be aware that discontinuation of epidural analgesia may be accompanied by an initial significant increase in pain intensity.

- Before stopping epidural analgesia, consider the following:

    1. The severity of pain and the epidural infusion rate.
    2. If the patient can tolerate other forms of analgesia. This should be prescribed regularly and as required.

- After discontinuing an epidural infusion, the epidural catheter can remain in-situ awhile. This allows for the option to re-establish the epidural should the systemic analgesia fail.

- Unlike systemic opioids, epidural doses cannot be used as a guide for the prescription of subsequent systemic analgesic regimes.

- If a patient is “nil by mouth” at the termination of epidural analgesia, IV PCA is an ideal step-down technique.

- Most patients convert to oral analgesia on cessation of an epidural. The epidural should continue until the patient is able to tolerate a minimum of 60 mls of fluid hourly, and able to absorb analgesia.

- Oral analgesia should only be given to patients with nasogastric tubes if spigotted or removed.

- The first doses of an alternative analgesia should be given immediately before the epidural infusion is stopped.

- Regular pain assessments must continue and providing that the patient remains comfortable after 3-4 hours, the epidural can be removed (Appendix 2 and Appendix 3).

- If the patient decides the epidural analgesia is no longer needed, full information should be provided regarding alternatives and emphases placed on their comparative efficacy.

26. Removal of the epidural

A trained and competent member of staff should remove the epidural catheter

- Before an epidural catheter is removed, it is essential to consider the clotting status of the patient's blood. If the patient is fully anticoagulated a clotting profile must be performed and advice sought from medical staff as to when the catheter can be removed if the patient is receiving prophylactic anticoagulants, the following guidelines are recommended (Hotlocker and Wedel 1998, Rowlingson and Hanson 2005, Vandermeulen et al, 1994) (Appendix 9).

- Low-dose low molecular weight heparin, if given once daily, the epidural catheter be removed at least 12 hours after the last injection and four hours prior to the next dose.

- If problems are encountered removing the epidural catheter contact the on-call Anaesthetist (Bleep 063) or the Acute Pain Service (Bleep 115).

- Once the catheter is removed, the condition of the catheter tip and the insertion site should be documented. If the catheter is broken or damaged the on-call Anaesthetist (Bleep 063) or the Acute Pain Service (Bleep 115).

- The total amount of epidural solution infused should be recorded on the appropriate documentation and any remaining controlled drug disposed in accordance with local policy.

27. Managing complications and side effects

Complications of epidural analgesia may be related to the epidural needle or catheter, the equipment, or side effects of drugs. Management of these complications is summarised in below:

Problem	Cause	Prevention	Action
Headache (Gaiser, 2013)	Dural Puncture	Expertise of practitioner inserting the epidural.	Bedrest: headaches will be less severe if the patient lies flat. Replacement fluids either intravenously or orally encourage formation of CSF. Administer analgesics for headaches. If the headache does not settle, contact the Anaesthetist who will consider and epidural blood patch to seal the puncture (Gaiser, 2013).
Sedation and respiratory depression (opioid toxicity) Circumoral tingling and numbness, twitching, convulsions and apnoea (local anaesthetic toxicity)	If the catheter migrates into a blood vessel, signs of opioid or local anaesthetic toxicity can occur	Expertise of practitioner inserting the epidural. Careful monitoring of the patient to detect early symptoms.	Stop the epidural infusion. Contact the Acute Pain Service, Anaesthetist or call for emergency assistance. Treat the patient for complications of opioid of local anaesthetic overdose. If needed support ventilation (Chapman and Day, 2001).

Problem	Cause	Prevention	Action
Apnoea, profound hypotension and unconsciousness	If an epidural catheter migrates from the epidural space into the intrathecal space to the CSF, the analgesic solution may reach as high as the cranial subarachnoid space. If this occurs the respiratory muscles are paralysed together with the cranial nerves, resulting in apnoea, profound hypotension and unconsciousness. This is because intrathecal doses are calculated as one-tenth of the epidural dose and migration from the epidural space to the intrathecal space leads to drug overdose	Expertise of practitioner inserting the epidural. Careful monitoring of the patient to detect early symptoms.	Stop the epidural infusion. Call for emergency assistance. Prepare emergency equipment to support respiration and ventilation. Prepare emergency drugs and intravenous fluid and administer as directed. Prepare equipment for intubation.
Back pain and tenderness and nerve root pain with sensory and motor weakness.	Epidural haematoma	Assessment of anticoagulation status before insertion and removal of the epidural. Avoiding multiple attempts.	Stop infusion without removing catheter. Urgent MRI Urgent neurological assessment (Horlocker, 2011)
Back pain and tenderness. May have red and purulent discharge from catheter exit site. May also develop nerve root signs with neuropathic pain and sensory/motor weakness.	Epidural abscess (Hebl and Niesen, 2011)	Maintain aseptic technique when assessing the epidural analgesic system. Monitor temperature regularly and check insertion site for evidence of infection. Increased in immunocompromised, IVDU, malignancy and diabetes	Urgent neurosurgical consultation. In absence of neurological signs treat with antobiotics If the epidural catheter is still *in situ,* remove and send tip off for culture and sensitivity. Decompression within 8 hours of onset maximises full recovery (ANZCA and FPM. 2015)

Problem	Cause	Prevention	Action
Headaches, fever, neck stiffness, photophobia	Meningitis	Maintain aseptic technique when Assessing the epidural analgesic system. Monitor regularly	Assist Doctor or Anaesthetist to obtain CSF sample for microbiology analysis. Initiate antibiotic therapy as per hospital policy. Non-pharmacological measures for symptom management e.g. dim lights. If the infection does not respond to antibiotics it may be necessary to remover the catheter and convert back to systemic analgesia until the meningitis is resolved (Mercadante, 1999)
Pain, paranaesthesia, numbness in lower extremities that may progress to paresis	Intrathecal catheter granuloma (an inflammatory mass that forms around the catheter tip	In long term therapy an intrathecal may be preferable to epidural to minimise the development of a granuloma due to the presence of the CSF	MRI scan to diagnose the problem. Stop infusion - granuloma may resolve on their own once the infusion is stopped (Du Penn 2005).

28. Concurrent antithrombotic and antiplatelet therapy

The following recommendations are not evidence based, as the incidence of dural hematomas is very small and therefore controlled studies are not possible. They are primarily based on the opinion and on consensus statements published in the United States (Horlocker et al, 2010, Horlocker, 2011) United Kingdom (Harrop-Griffiths et al., 2013), and Europe (Gogarten et al., 2010). However, they cannot replace an individual risk-benefit analysis for every patient.

Warfarin

- A major difficulty in making recommendations is the wide interindividual variability in response to this drug. In patients on long-term warfarin therapy, haemostasis may require 3-5 days to normalise once the drug has stopped.

- Coagulation status the prothrombin time (PT) and international normalised ratio (INR) should be checked before the insertion of an epidural needle. An INR <1.5 is regarded as relatively safe in this setting.

- Some patients are prescribed low-dose warfarin for postoperative thromboprophylaxis (e.g. after major surgery). The best time for removal of an epidural catheter in these patients is not known. Relying on an average time to effect is 48 hours (peak effect 3-5 days) may by dangerous in individual patients, as there will be an effect on the INR after a single dose of warfarin in some.

- If the epidural catheter is left longer than 24 hours after the first dose after the first dose, monitoring of coagulation is suggested. An INR ideally <1.5 at time of removal.

Heparin

- Epidural catheterisation appears to be relatively safe in patients who receive intravenous heparin, either during or after surgery. However, consideration must be given to both the timing of both catheter placement and catheter removal *before* heparin is given and *after* heparin is commenced. The current guidelines suggest that after placement of epidural needles and catheter, administration of IV heparin should be delayed for at least an hour.

- If a postoperative heparin infusion is required, the epidural catheter should be removed only after the infusion has been suspended for 2-4 hours and after the patient's coagulation status has normalised, as indicated by measurement of the activated partial thromboplastin time (APTT).

Standard unfractionated heparin (subcutaneous)

- Low-dose standard heparin is commonly administered for thromboprophylaxis, although regarded as insufficient after orthopaedic surgery, and is generally considered safe to use in patients with concurrent epidural analgesia at doses under 10,000 units daily. However appropriate precautions should be taken regarding the timing or epidural catheter insertion and removal, as the peak effect of the dose of standard heparin is likely to be seen 1-2 hours and the duration of the effect may be 4-6hours or more.

- Some guidelines recommend that epidural catheters can be inserted or removed 1-2 hours before the next dose in a twice daily (BD) regimen (commonly 5000 units BD).

- Longer therapy with heparin can result in heparin-induced thrombocytopenia. Guidelines recommend a platelet count prior to insertion or removal, after more than 5 days of heparin therapy.

29. Newer Anticoagulants

Dabigatran

- An oral anticoagulant used primarily in patients with atrial fabulation. There are significant concerns about the safety of epidural techniques. Difficulties in the reversal and prolonged duration of effect (longer in renal impairment) create uncertainty about its safe use. Dabigatran should be discontinued 7 days before an epidural and not removed 6 hours before the drug is restarted.

Rivaroxaban

- An oral antithrombotic recommended for thromboprophylaxis after total hip or knee joint replacement. It also has a long half-life (9hours), increased in renal impairment, but its effects can be monitored by measuring PT and APTT. Guidelines recommend about a 24-hour interval between the last dose or rivaroxaban in patients with normal renal function, with this interval being extended in patients with renal impairment. Administration in patients with an epidural catheter *in situ* is not recommended.

30. Non-steroidal anti-inflammatory drug and other antiplatelet

Non-steroidal anti-inflammatory

- The use of NSAIDS including aspirin (given alone) has not been identified as an independent risk factor for epidural haematoma.

- However, concurrent use with other medications that can affect coagulation status (e.g. heparin or LMWH) may increase the risk. Therefore, if required in combination with other anticoagulants, coxibs (selective COX-2 inhibitors) should be used in preference to nonselective NSAIDs.

Other antiplatelet drugs

- The recommended time intervals for other antiplatelet agents is dependent on the duration of effect. Suggested acceptable intervals between cessation of the drug and block are:

Clopidogrel and prasugrel	7 days
Ticlopidine	10-14 days
Eptifibatide and tirofiban	8 hours
Abciximab	48 hours

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Appendix 1. Care instructions following an epidural

GIG CYMRU NHS WALES
Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

Care instructions following your epidural

Following your epidural, it is important to know the signs you need to be aware of. The following will tell you what to look out for and what actions to take if you think you have a problem.

Although rare (1 in 10,000) it is possible that serious complications from having epidural analgesia could occur. The epidural needle is inserted into the epidural space which is close to the spinal cord, a collection of pus or a blood clot can cause pressure. In the unlikely event of this happening it is crucial to diagnose and treat urgently.

- To minimise any risk, it I important for you to be aware of any signs or symptoms you could experience requiring urgent treatment.
- Redness, pus, tenderness or pain where the epidural was inserted.
- Feeling generally unwell even though all seems well with the surgical wound.
- High temperature or neck stiffness.
- Numbness or weakness in your legs or an inability to stand.
- Difficulty in passing urine or loss of bowel control.

If you have any of the above symptoms you must attend the Emergency Department of the hospital immediately and ask them to inform the Anaesthetist. Please bring this letter with you.

Acute Pain Service, Ysbyty Gwynedd May 2019.

Appendix 2. Epidural Monitoring Chart

Patient label	Anaesthetist:
	Insertion (date & time):
	Ward:
	Site of insertion:
	Other attempted sites:
	Catheter tip to skin: cms
	Bloody tap?
Catheter dose pre-catheter insertion (date & time):	
72 hours of epidural infusion commended (date & time)	
Discuss removal and alternative analgesia with an Anaesthetist or the Acute Pain Service	
Assess the anticoagulation status before removal, see below	
Notes or instructions:	

Date	Time	Area of sensory block		Degree of motor block		Pain Score	Epidural site inspection
		Right	Left	Right	Left		

For full instructions refer to the hospital Epidural Policy

Anticoagulant status: APTT _____ INR _____

Planned catheter removal: Date _____Time _____

Catheter removed: Date _____Time _____ Signature: _____

Previous clexane dose: Date _____Time _____(12 hrs max)

Next clexane dose post catheter removal: Date _____Time _____(4 hrs max)

Care instruction sheet given on discharge: Yes/ No

Appendix 3. Epidural Observation Monitoring Chart

Epidural site inspection:

- **Twice a day:** for swelling, inflammation and leakage.
- **48 hours:** following the removal of catheter

LEVEL OF SENSORY BLOCK	
The level of the block should be measured every **8 hours** or **15 minutes** following a bolus or change of infusion	
T2	Clavicle
T4	Nipple
T6	Xiphisternum
T10	Umbilicus
T12	Groin
• If the level is greater the T4 STOP infusion and inform the Acute Pain Team (Bleep: 115) or Anaesthetist on-call (Bleep 063).	

MOTOR BLOCK	
Use the Bromage scoring system: • 1 hourly for the first hour • 2 hourly for the next 24 hours	
0	Flexion of knees and feet
1	Just able to move knees and feet
2	Able to move feet only
3	Unable to move feet and knees
• Contact the Anaesthetist **urgently** if patient is unable to move both legs • Contact the Anaesthetist is motor lock is heavy i.e. 2 or getting worse	

PAIN SCORE	
0	No pain
1	Mild pain
2	Moderate pain
3	Severe pain
• Asses pain on movement or deep breathing • Seek help from the Anaesthetist or Acute Pain Team, if pain scores 2 or 3.	

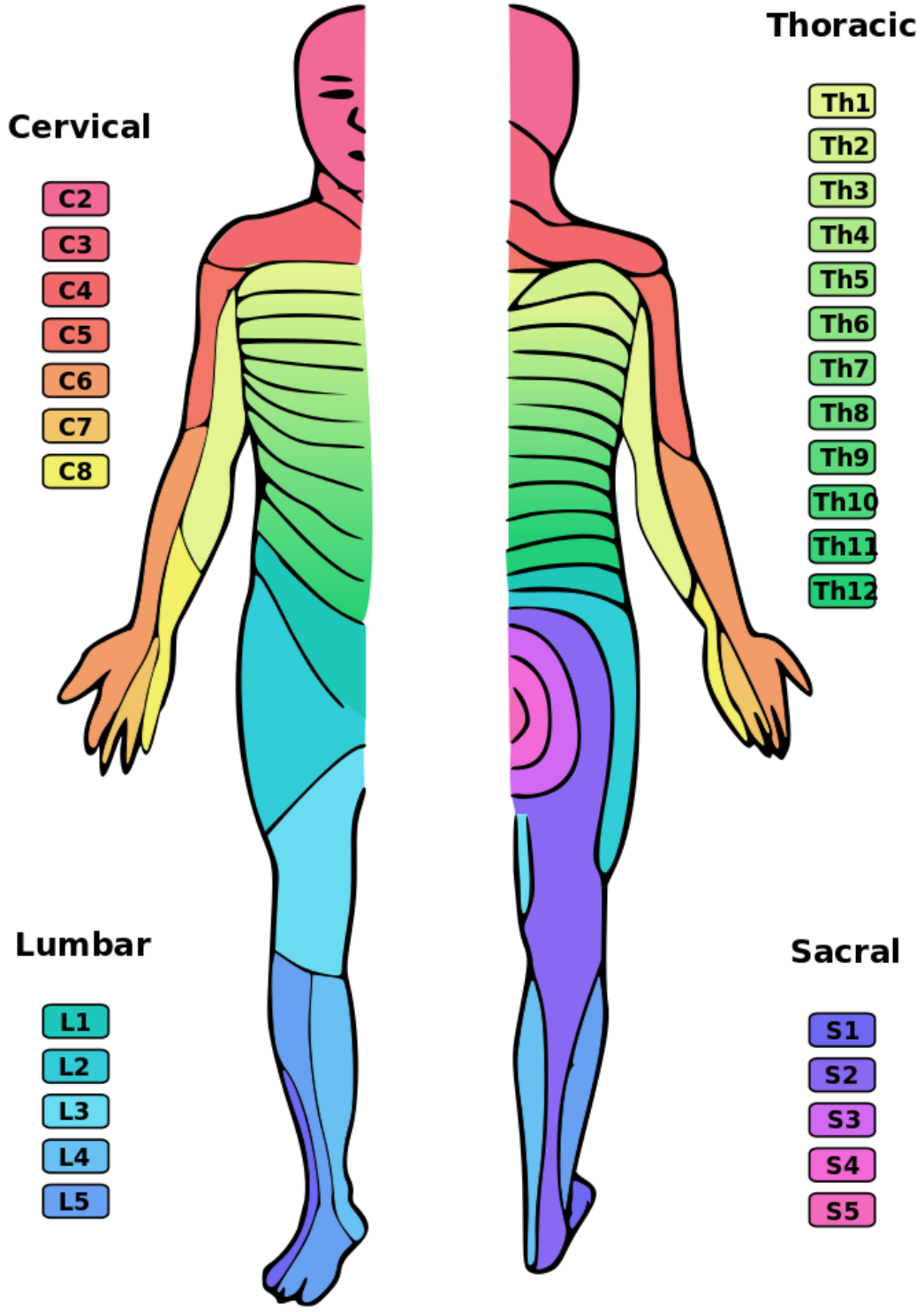
BLADDER
Most patients will have a urine catheter as an epidural infusion of local anaesthetic causes painless retention of urine.

CATHETER REMOVAL
• The epidural catheter can only be removed **12 hours**, after the last clexane dose. • If a bloody tap or blood in the catheter, this increases to 24 hours on the advice of the Anaesthetist. • Other coagulation abnormalities should also be considered. • Following insertion of removal of an epidural catheter no clexane for **4 hours**. • Do not warfarin whilst the epidural catheter is in place • **Ensure motor block assessment is continued 2 hourly for the next 12 hours**

Appendix 4. Epidural exit site dressing change

Essential Equipment	
• Sterile dressing pack including gloves • Skin cleaning agent, e.g. chlorhexidine in 70% alcohol • Transparent occlusive dressing	
Pre-procedure	
Action	**Rationale**
1. Explain and discuss the procedure with the patient.	To ensure the patient understands the procedure and gives valid consent (NMC, 2013).
2. Wash hands with soap and water. Clean trolley with chlorhexidine in 70% alcohol with a paper towel.	To minimise cross infection (Preston, 2005).
3. Position patient comfortably on their side or sitting forward so the site is easily accessed without undue exposure of the patient.	To maintain patient dignity and comfort (Chapman and Day, 2001).
4. Prepare trolley with sterile field and cleaning solution	To minimise the risk of infection and ensure equipment is available (Preston, 2005).
Procedure	
5. Remove the old dressing and place in a disposable bag.	To prevent cross-infection (Fraise and Bradley, 2009).
6. Wash hands with bacterial hand scrub. Put on apron and gloves.	To minimise the risk of microbial contamination (Fraise and Bradley, 2009).
7. Observe the site for any signs of infection such as redness, swelling or purulent discharge. If any of these are present, contact the Anaesthetist or Pain Team for advice.	To ensure careful monitoring of the site to minimise the chance of any infection (Royal College of Anaesthetics, 2004).
8. Clean site with skin-cleaning agent (chlorhexidine in 70% alcohol).	To minimise the risk of infection (Hebl, 2006).
9. Apply transparent occlusive dressing over the whole area.	To anchor, minimise infection and allow observation of the site (Royse el al. 2006).
Post procedure	
10. Remove gloves and apron and dispose in clinical waste.	To prevent environmental contamination (Preston, 2005).
11. Wash hands with soap and water.	To reduce cross-infection (Preston, 2005).

Appendix 5. - Dermatomes



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Appendix 6. Epidural sensory block assessment

Essential Equipment	
• Ice or ethanol spray • Copy of the dermatome distribution • Documentation	
Pre-procedure	
Action	**Rationale**
1. Explain and discuss the procedure with the patient. Obtain informed verbal consent from the patient	To ensure the patient understands the procedure and gives their valid consent and to ensure the patient has time to assess information and ask questions (NMC, 2018)
Procedure	
2. Explain to the patient that they need to report: • If the temperature of the ice/cold changes or becomes warmer. • If they cannot feel the ice/cold sensation at all	This will indicate the dermatome level at which the epidural analgesia is working
Action	
1. Remove any clothing that may restrict assessment. 2. Test the ice/ cold spray on the area of the body that should not be affected by the epidural infusion (e.g. back of hand). 3. Starting with the top of the chest (just above nipple level). Place the ice/cold spray above the incision or level of pain and ask the patient if this feels as cold as when placed on the test area. Continue this procedure down the torso and legs and the legs is appropriate for the level of epidural analgesia and compare the right and left of the body	The ice spray needs to be applied directly o the skin to undertake the assessment To ensure the patient and feel the cold sensation
Post procedure	
6. Document at what level the patient can detect a change as per dermatome	

Appendix 7. – Management of inadequate analgesia

Management of inadequate analgesia

Reassess the patient

- Consider another cause for new or increased pain such as the development of postsurgical or postinjury complication, for example, bowel perforation and/or peritonitis after abdominal surgery, compartment syndrome after orthopaedic surgery or limb surgery, or the pain is not covered by epidural analgesia.

- Test the level of the block

Bilateral block but inadequate spread

- Give a bolus dose and increase the rate of infusion

Unilateral block

- Suggest the epidural tip may have exited the epidural space through and intervertebral foramen or there is an anatomical reason for asymmetrical spread

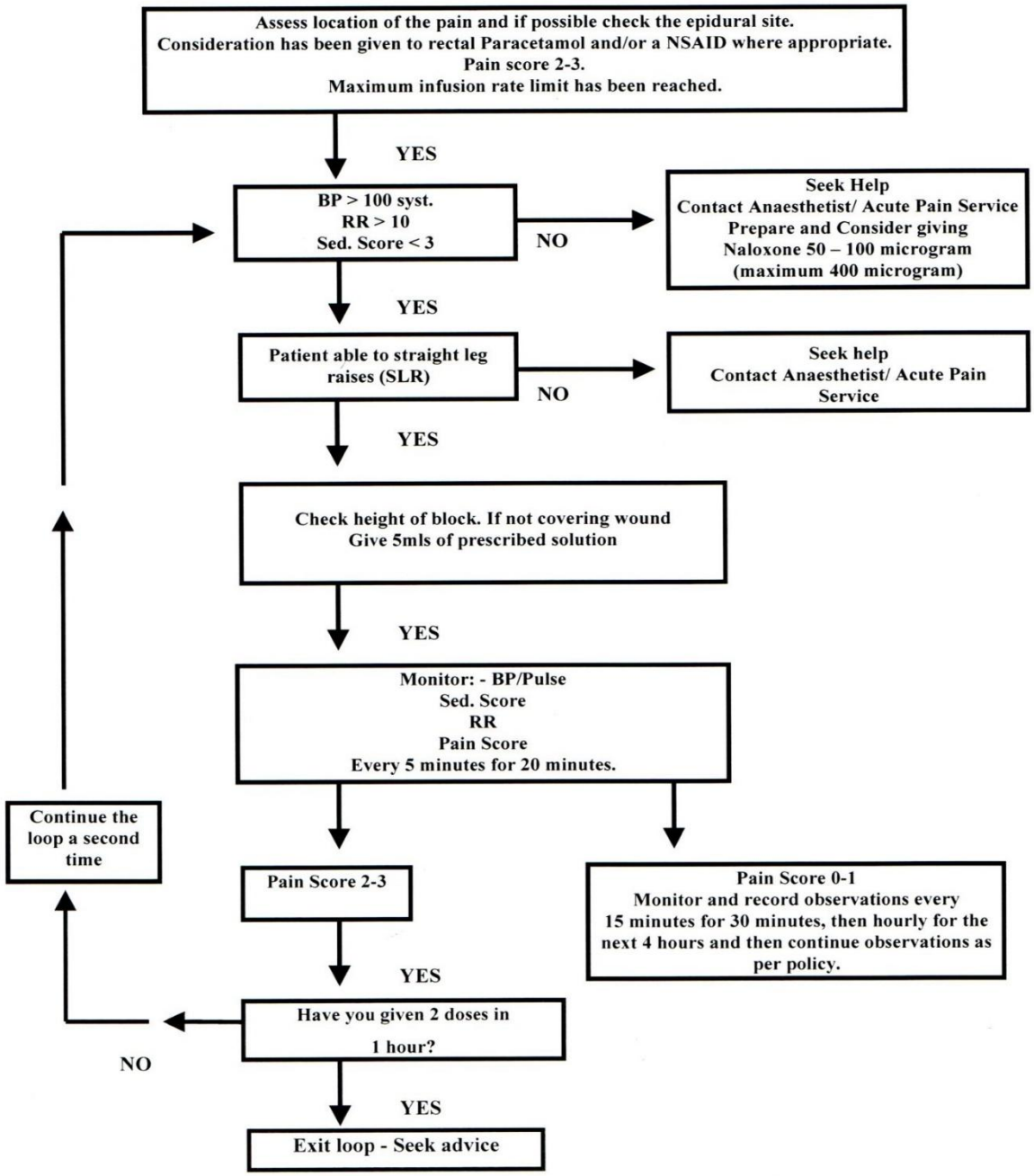
- Try a larger bolus (Anaesthetist only) and/or

- Withdraw the catheter a little

No block or generally poor pain relief

- Exclude intravascular catheter migration by aspiration test (Anaesthetists only)
- Check position of the catheter using a “test dose” of 3-8 mL of local anaesthetic solution (e.g. 1% lidocaine) and test the level of the sensory block (Anaesthetist only)
- Is the “test dose “shows no block the catheter is displaced, prescribe alternative analgesia or reinsert a catheter (allow adequate time after any heparin administration) (Macintyre and Schug, 2015).

Appendix 8. Epidural top up flow chart for unrelieved pain



Appendix 9. Epidural catheter removal

Essential Equipment	
• Sterile dressing pack including gloves • Skin cleaning agent, e.g. chlorhexidine in 70% alcohol • Specimen container (if epidural catheter needs to be sent for bacterial culture • Occlusive dressing • Alcohol hand rub	
Pre-procedure	
Action	**Rationale**
1. Explain and discuss the procedure with the patient.	To ensure the patient understands the procedure and gives valid consent (NMC, 2018).
2. Wash hands with soap and water. Clean trolley with chlorhexidine in 70% alcohol with a paper towel.	To minimise cross infection (Preston, 2005).
3. Open dressing pack.	To provide a clean working surface (Parker, 2004).
Procedure	
4. Wash hands and remove tape and dressing from the catheter. Put on apron and gloves.	To minimise cross infection (Preston, 2005).
5. Wash hand with bacterial hand scrub. Gently, in one swift moment, remove the catheter. Check the catheter is intact, by observing the tip of the catheter is marked blue and that the 1cm markings along the length of the catheter are intact.	To minimise the risk of microbial contamination (Fraise and Bradley, 2009). To ensure the catheter is removed intact with minimum discomfort to the patient.
6. Clean around the catheter exit site using skin cleaning agent.	To minimise contamination of the site by micro-organisms.
7. Apply an occlusive dressing and leave it *in-situ* for 24 hours.	To prevent inadvertent access of micro-organisms along the tract.
Post procedure	
8. The epidural tip may be sent for culture and sensitivity if infection is suspected, or according to local policy.	To prevent environmental contamination (Preston, 2005).
9. Remove gloves and apron and dispose. Wash hands with soap and water.	
10. Document that the catheter was removed intact in nursing notes	